Statistics in Health Sciences

Assignment 1

Cluster analysis with missing data using the `R` packages `mice` and `miclust`

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1 Introduction

The aim of this assignment is to apply the framework proposed by Basagaña et al. to the data set `copd` in the file `copd_redux.RData`. Data `copd` is a subset of 45 variables of the original data analyzed in the cited paper.

2 Analysis

2.1 Description of missingness

Provide the same missingness description than in the original analysis. Specifically, reproduce the following text that is shown in the original paper:

The required packages are:

- Number of participants;
- Number of variables;
- Missing data percentage of each variable;
- Missing data percentage of less than 5% of missing;
- Percentage of variables with all variables without any missing;
- Percentage among all cell with missing data.

```
> load("copd_redux.RData")
> n <- length(copd$id) #participant number
> v <- ncol(copd) #variables number
> M <- n*v #cell number
>
> missing <- colSums(is.na(copd))
> miss_percentatge <- (missing/n)*100 #missing
>
> minimum_missing_pp <- min(miss_percentatge) #minimum missing percentage
> maximum_missing_pp <- max(miss_percentatge) #maximum missing percentage
>
> miss_under_5 <- sum(miss_percentatge < 5)
> pctg_under_5 <- (miss_under_5/v) * 100
> #percentage of variables with less than 5% missing
>
> complete <- sum(miss_percentatge == 0)
> complete_pctg <- (complete/v)*100
> #percentage of patients with complete data on all variables
>
> overall <- sum(is.na(copd))
> overall_pctg <- (overall/M)*100
> #Overall percentage of the information is missing
```

Now we complete the missing values of the text:

"The subset used here included 342 participants and 45 variables. The range of missing data in the different variables went from 45%, with 0% to 31.6%, with 68.9% of the variables having less than 5% of the data missing. Only 11.1 of the participants had complete data on all 45 variables. Overall, 11.1 of the information was missing; that is, 5.4% of 15,390 cells ($45 \times 342 = 15,390$) had missing data."

2.2 Multiple imputation

Apply multiple imputation to `copd` data using 100 imputations. Explain in a reasoned manner how you decided to set the arguments `method` and `predictorMatrix` when using the `mice()` function.

Before staring to decide the arguments for `method` and `predictorMatrix` we should do a little summary to know the classes of each variable of this `copd` data-set.

```
> str(copd)

## 'data.frame': 342 obs. of  45 variables:
##  $ id                : num  3045 5155 5062 5146 1055 ...
##  $ index_barthel2v1  : num  100 100 100 100 100 90 100 100 100 100 ...
##  $ quo_calcpostv1    : num  40.5 55.2 28.6 67 40.5 ...
##  $ rv_prepv1         : num  180 134 202 125 166 ...
##  $ incp_fevv1        : num  10.619 -0.694 -4.255 -1.408 20.976 ...
##  $ fevp_postv1       : num  40 76.9 20.1 70.4 40.7 ...
##  $ rv_tlcv1          : num  54.8 46.5 71.8 46.2 51.8 ...
##  $ ctnf_s_mv1        : num  2.957 0.045 1.023 0.64 0.557 ...
##  $ vtdvdbid_ecov1    : num  30 32 NA 29 NA NA NA NA NA NA ...
##  $ pp_ecov1          : num  11 8 10 10 9.5 9 10 11 9 9 ...
##  $ sin11_v1          : Factor w/ 2 levels "0","1": 2 1 1 2 1 2 1 2 1 1 ...
##  $ vas2_v1           : num  8 4 0 4 5 ...
##  $ ic_prepv1         : num  67.9 96.7 38.7 80.2 85.2 ...
##  $ iam_chv1          : Factor w/ 2 levels "0","1": 1 1 1 1 1 1 1 2 1 1 ...
##  $ sin15_v1          : Factor w/ 2 levels "0","1": 2 1 1 1 2 1 1 1 2 1 ...
##  $ cil8_s_mv1        : num  3.59 2.75 3.72 2.8 6.04 ...
##  $ sin7_v1           : Factor w/ 2 levels "0","1": 1 1 1 2 1 1 1 1 1 1 ...
##  $ sin8_v1           : Factor w/ 2 levels "0","1": 2 2 2 2 2 2 1 1 1 2 ...
##  $ ch_v1             : num  0.4 1 2.6 1.8 0.8 ...
##  $ vc_prepv1         : num  73.4 92.4 49 88.7 73.4 ...
##  $ dtdve_ecov1       : num  63 45 55 60 53 48 49 51 57 49 ...
##  $ cccs1_v0          : Factor w/ 2 levels "0","1": 2 1 1 1 2 1 1 1 1 2 ...
##  $ kcals_set_v0      : num  1635 15055 2265 6054 10965 ...
##  $ impactsv1         : num  29.46 3.55 35.51 24.72 23.22 ...
##  $ monoA_v1          : num  0.41 0.495 0.827 0.543 0.289 ...
##  $ lvmass_ase_ecov1  : num  302914 142108 219991 287863 199797 ...
##  $ vtit_catv1        : Factor w/ 2 levels "0","1": 1 NA 1 2 2 NA NA 1 1 NA ...
##  $ urg_ambv0         : Factor w/ 2 levels "0","1": 1 1 1 1 2 1 1 1 2 1 ...
##  $ po2_v1            : num  63.1 80.5 63.5 92 80 ...
##  $ ahpf_v1           : num  2.5 3.2 3 2.9 4 ...
##  $ pro_v1            : num  6.73 6.93 7.7 6.8 6.9 ...
##  $ basA_v1           : num  0.1104 0.0228 0.0165 0.032 0.0346 ...
##  $ fc_pmv1           : num  0 2 0 1 0 1 5 0 0 0 ...
##  $ tad_v1            : num  80 93 80 90 75 70 94 90 80 86 ...
##  $ col_v1            : num  174 282 NA 174 145 135 168 213 211 212 ...
##  $ sin5_v1           : Factor w/ 2 levels "0","1": 2 2 1 1 2 1 2 1 1 1 ...
##  $ bc_v1             : num  0 0 0 0 0 0 0 0 0 0 ...
##  $ fvcp_postv1       : num  66.1 102.9 43.2 83.5 67.2 ...
##  $ vas1_v1           : num  1 2 1.7 3.2 0 ...
##  $ sin6_v1           : Factor w/ 2 levels "0","1": 2 1 2 2 1 1 1 1 2 2 ...
##  $ graudisn_v1       : Factor w/ 6 levels "0","1","2","3",..: 4 3 3 4 3 6 6 3 3 1 ...
##  $ cpcr_mg_l_mediav1: num  0.08 3.11 1.07 0.08 0.08 ...
##  $ neutA_v1          : num  4.78 6.05 5.22 3.51 2.74 ...
##  $ ahpl_v1           : num  1 2.2 2.3 5.2 1 ...
##  $ activityv1        : num  61.3 35.5 41.8 35.6 36.3 ...
```

As we see, most of the variables are numerical, however there are some variables which are factors. Their names and corresponding numbers of levels are listed below:

```
> categoricas <- names(copd)[sapply(copd, is.factor)]
> categoricas_latex <- gsub("_", "\\_", categoricas)
> niveles <- sapply(copd[categoricas], nlevels)
```

```
> binarias <- names(niveles[niveles == 2])
> binarias_latex <- gsub("_", "\\\\_", binarias)
> no_binarias <- names(niveles[niveles > 2])
```

As we can see, this data set has 11 variables that are factors, and their names are `sin11_v1, iam_chv1, sin15_v1, sin7_v1, sin8_v1, cccs1_v0, vtit_catv1, urg_ambv0, sin5_v1, sin6_v1, graudisn_v1`. The following factors are binary distributed: `sin11_v1, iam_chv1, sin15_v1, sin7_v1, sin8_v1, cccs1_v0, vtit_catv1, urg_ambv0, sin5_v1, sin6_v1`